

# Özgür Kurul | Software Team Lead & Architect

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Languages: Turkish (Native), English (Advanced/Fluent)

## Professional Summary

Results-driven **Software Team Lead and Architect** with over 10 years of experience designing, scaling, and modernizing high-volume, mission-critical systems. Proven expertise in directing the end-to-end transformation of legacy monoliths into robust, containerized **Microservices** architectures. Specializes in building comprehensive **IoT ecosystems** and real-time data streaming pipelines utilizing Event-Driven Architecture (**Kafka, Debezium**). Adept at leading cross-functional Agile teams to deliver highly available, cross-platform solutions (.NET, React and React Native) integrating GIS, AI, and complex hardware protocols. Passionate about driving technical excellence, establishing sustainable CI/CD cultures, and architecting scalable backend solutions that power real-world, high-throughput operations.

## Experience

**Drivee Technology | Software Team Lead (Cross-Platform)** Istanbul, Turkey | February 2024 - Present

- **Team Leadership & Agile Management:** Lead and mentor a cross-platform team, managing the end-to-end SDLC for Backend, Web, and Mobile applications. Established Agile/Sprint methodologies, conducted 1-on-1s and performance reviews, and enforced code quality using SonarQube and GitHub Copilot Review.
- **Architecture & Transformation:** Directed the complete transformation of a legacy portal into a scalable, containerized microservices architecture (Web API, Producer/Consumer) with Web (React.js) and Mobile (React Native) clients. Applied advanced patterns including Clean Architecture, Layered Architecture, Vertical Slicing, API Gateway, and the Outbox Pattern.
- **IoT Ecosystem & Global Vendor Management:** Evolved the core platform into a comprehensive IoT ecosystem. Managed **technical communications** and product sourcing with international hardware vendors to successfully integrate diverse vehicle tracking devices (Teltonika, JT808, Armoli, RFID, CAN bus, Sensors) and camera systems (CMSv6, Jimi TrackSolidPro). Also integrated 3rd-party fuel providers (like Total, Petrol Ofisi) into the system.
- **Telecommunications & Mission-Critical Alerting (Cold Chain):** Developed a high-availability Cold Chain tracking system with automated rule-based alerts to ensure food safety for major franchises (KFC, HD İskender, Pidem, Dürümle). Engineered robust communication gateways, successfully integrating bidirectional SMS pipelines, automated voice calls, and direct call center routing to enable real-time, rule-based incident response and direct user engagement.
- **AI & Real-Time Data:** Integrated Gemini AI to create "Drivee AI" for predictive corporate fleet analysis. Engineered loose-coupled, event-driven data pipelines using Kafka and Debezium for real-time synchronization.
- **GIS & Mapping:** Implemented advanced routing and geofencing systems utilizing Google Services, OpenStreetMap (Overpass, Nominatim API), PostGIS, and NetTopologySuite.
- **Infrastructure & DevOps:** Ensured database geo-redundancy with MS SQL Server Always On and established Master/Replica architectures for Redis and MongoDB. Automated CI/CD pipelines via GoCD, GitHub Actions, and Expo (automated TestFlight/App Tester distribution), while securing the system with WAF/CDN and monitoring via Elastic APM and Elasticsearch.

**Halk Bank (via Roofstacks) | Senior Backend Engineer** Istanbul, Turkey | October 2023 - February 2024

- **Mission-Critical Banking Systems:** Engineered highly secure backend microservices and architected the primary API Gateway for the bank's innovative "Child and Campus Banking" mobile application.
- **Security & Compliance:** Developed high-performance, resilient APIs to process sensitive financial transactions, strictly adhering to enterprise-grade banking security and compliance standards.

**Delivery Hero Tech Hub | Senior Software Engineer** Istanbul, Turkey | April 2022 - October 2023

- **Global Integration & Cross-Border Collaboration:** Contributed to the Integration Hub by designing seamless 3rd-party vendor integrations using .NET and AWS (SQS, DynamoDB, Lambda). Successfully collaborated with distributed, **international engineering teams** in a fully English-speaking environment to align technical requirements and drive global processes.
- **Developer Experience (DX) & Testing:** Built a comprehensive mock testing framework from scratch for 3rd-party integrators, significantly reducing integration lead times and accelerating the global partner onboarding process.

## Yemeksepeti | Senior Software Engineer

Istanbul, Turkey | August 2021 - April 2022

- **High-Traffic E-Commerce:** Played a pivotal role in the backend development and successful launch of the "Yemeksepeti Mahalle" quick-commerce platform.
- **Event-Driven Order Lifecycle:** Designed, developed, and maintained robust microservices handling the end-to-end order lifecycle (from placement to real-time delivery) utilizing .NET, PostgreSQL, and AWS SQS/SNS under heavy user traffic.

## Defacto | Software Engineer

Istanbul, Turkey | August 2020 - August 2021

- **Omnichannel Operations:** Developed core features for the global Omnichannel e-commerce project, streamlining processes from online order management to automated courier assignment and return logistics.
- Enhanced core e-commerce functionalities by optimizing 3rd-party integrations (Marketplace, shipping, and courier), resulting in improved overall system efficiency.

## Novartz | Software Engineer

Adana, Turkey | January 2016 - July 2020

- **Foundational IoT Architecture:** Acted as the core Project Leader for the "Drivee" live vehicle tracking system establishing the initial architecture for real-time tracking, geofencing, and complex driver scoring algorithms using high-performance TCP/IP Socket programming and real-time data streaming.
- **Data Analytics & Early IoT:** Developed an automated data extraction and reporting system for Starbucks vending machines (Shaya Group), laying early foundations for large-scale IoT data analytics.
- **Enterprise CRM & Portal Solutions:** Designed and delivered over 15 customized CRM, CMS, and corporate portal systems for diverse B2B clients, successfully implementing complex 3rd-party integrations (Banking Payment gateways, ERP systems like LOGO and E-Flow), streamlining their internal operations and data workflows.
- **E-Commerce & B2B Solutions:** Led the end-to-end development of the MetalAVM e-commerce platform and delivered highly customized CRM portal solutions for diverse corporate clients.

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## Skills

- **Leadership & Methodology:** Agile, Scrum, Jira, Performance Management, 1-on-1 Mentoring, System Architecture.
- **Architecture Patterns:** Microservices, API Gateway, Outbox Pattern, Event-Driven/Async Messaging, Clean Architecture, Vertical Slicing, Cache Strategies for Distributed Systems.
- **Backend:** C#, .NET 8, Web API, REST, Dapper, EF, SignalR, Hangfire, Quartz.
- **Frontend & Mobile:** React Native, React.js, JavaScript, TypeScript, Expo.
- **Databases & GIS:** MS SQL, PostgreSQL, PostGIS, NetTopologySuite, Redis, MongoDB.
- **Messaging, IoT & AI:** Kafka, RabbitMQ, Debezium, N8N, Gemini AI Integration, AWS SNS/SQS, Lambda, Device Protocols (JT808, CMT6, Teltonika, CAN bus, RFID, Sensors), IVR Calling, SMS Send and Tracking.
- **DevOps, Cloud & Security:** Docker, Kubernetes, CI/CD Strategy with GitHub Actions and GoCD, WAF, CDN, Firebase, Nginx, Kong, HAProxy, SonarQube, GitHub Copilot Review.
- **Monitoring & Mapping:** Elastic APM, Elasticsearch Log Trace, Graylog, Health Checks, Google Maps APIs, OpenStreetMap (Nominatim).

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## Education

**Cukurova University** | Adana, Turkey | 2012 - 2016

*Bachelor of Engineering (B.E.), Computer Science*

**AGH University of Science and Technology** | Kraków, Poland | 2015

*B.E., Computer Science (Erasmus+ Exchange Program)*

**Osmaniye Korkut Ata University** | Osmaniye, Turkey | 2010 - 2012

*Foundation Degree, Computer Programming*

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## Courses & Certifications

- Cybersecurity Awareness (8 Hours-2023): Phishing Attacks, Malware Explained, The Internet of Things (IoT).
- Secure Code Warrior (8 Hours-2022 and 2023): Secure code camping.
- BGA Cyber Security Training (48 Hours-2019): Comprehensive training covering Penetration Testing, DDOS mitigation, Web Application Security, and Vulnerability Analysis.
- Innovation Management Training (24 Hours-2020): Advanced modeling and workshop applications for corporate innovation strategies.
- GDPR Compliance (2024)